DEVOPS ADVANCED PROFESSIONAL CERTIFICATION · DAPC

Session 6

Tests, TDD, continuous integration and trunk-based development

TODAY

Six blocks, 240 minutes

BLOCK	MINUTES
Work trail from session 5	15
Create skill to code a ticket using AI	20
Theory with Claude, in pairs	40
Tickets: three bugs, test first, then fix	90
The red build: discussion	35
Wrap-up and buffer	20

Create skill to code a ticket using AI

1	**name**	A short identifier for the skill
2	**description**	What it does and when to use it. This triggers the skill
3	**instructions**	The steps Claude follows once the skill is loaded
4	**allowed-tools** optional	Limits which tools Claude may use while the skill runs
	supporting	scripts/ for code, references/ for longer docs

OPENING

What the session 5 work trail shows

Linked PR

Every change has a PR that names its SCRUM ticket.

CI green

The checks ran and passed before review.

Classmate review

A second person read the change and commented.

Pick two PRs from different people. Ask: where did CI help, and what took longest?

WHY TESTS COME FIRST

CASE 1 · BANKING · KNIGHT CAPITAL, 2012

One of 8 servers still ran old code.

No automated check compared them. In about 45 minutes the firm lost roughly $440 million.



THEORY WITH CLAUDE

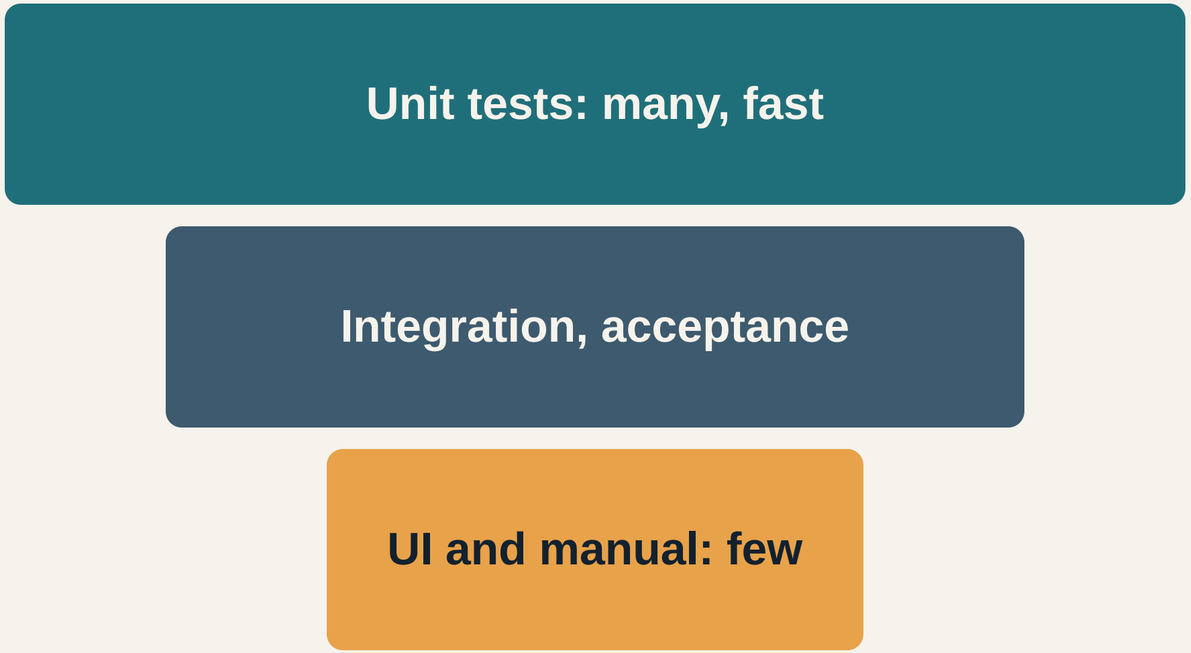
Three pairs, one topic each

Team A	Team B	Team C
Test types and the pyramid	TDD and fast, reliable tests	Integrate often: official trunk-based model vs develop flow, and the Andon cord
pp. 48 - 50	pp. 51-52	pp. 53-54

Five minutes each. Check against the official pages. One example from the Summit Cafe API.

TEST TYPES

Keep the base wide



- Official categories: unit, integration, acceptance.
- The ice-cream cone is the anti-pattern
- Strong coupling makes unit tests hard to write.

FAST AND RELIABLE

A few reliable tests beat many flaky ones

Reliable and small

Usually better than many unreliable tests. Start small and add over time.

p. 48

Fast enough to run

Fast tests get run. Slow or flaky tests get ignored. Run in parallel when needed.

FIND ERRORS EARLY

An API test found the bug? Add a unit test.

1

Test finds it

An acceptance or integration test fails. Feedback is slow.

→

2

Fix the bug

The code is corrected and the failing test passes.

→

3

Add a unit test

One that would have caught it earlier, faster and cheaper.

TEST-DRIVEN DEVELOPMENT

Red, green, refactor

Red

Write a test for the next behaviour. Watch it fail.

Green

Write the smallest code that makes it pass.

Refactor

Clean up the new and old code. Keep the tests green.

Kent Beck, Extreme Programming (1990)

CONTINUOUS INTEGRATION

The pipeline checks every change

AUTOMATIC

Unit test

AUTOMATIC

Integration

MANUAL

Exploratory · UAT · Staging · Production

Our **Checks** job is the automatic stage. Merging to develop is part of daily work.

TRUNK-BASED DEVELOPMENT

Long branches make merging painful

Long-lived branches

Integration gets exponentially harder the longer it waits.

Painful merges happen less often, so the next one is worse.

Official trunk-based practice

Commit to the trunk at least once a day.

Blocked commits: merge only if all automated tests pass.

OFFICIAL DEFINITION OF DONE

At the end of each interval we have…

Code that is integrated and tested

Working, and potentially deliverable

Shown in an environment like production

THE ANDON CORD

When the build breaks, no new work enters until it is fixed.

If someone needs help, bring in the help they need. A virtual Andon cord, like the one in the Toyota Production System.

DISCUSSION · 35 MINUTES

The red build

1. Who may stop the line, and how do they say so?

2. What counts as fixed: a green build, or a test that covers the bug?

SHIFT LEFT

Quality starts in planning, not in testing

Attention to quality should move earlier: to plan, design and develop, not only to test and release.

From session 5

Where did the problems in our PRs show up first?

TICKETS · TEST FIRST, THEN FIX

Three bugs, one test then one fix each

TICKETS	BUG
SCRUM-15 → 18	Order total ignores quantity: test first, then multiply price by quantity
SCRUM-19 → 20	Boolean quantity accepted (true counts as 1): test first, then reject it
SCRUM-21 → 22	Boolean menu id accepted (true orders Espresso): test first, then reject it

DEFINITION OF DONE · OUR COURSE

Your work trail depends on these four

1. A test is written before the code (ticket b)
2. CI is green: ruff and pytest pass
3. The ticket number is in the branch, commit and PR title

FIRST SKILL

Build your skill

A skill is a folder with instructions that Claude loads when a task matches.

Review a PR #28 for test quality

Create an skill for review MR and report if it has unit tests

WRAP-UP

Session 6 outcomes

- Tests that block a bad change
- One feature built test-first
- One CI check you proved blocks a merge